

Explaining the Relative Nature of General and Special Relativity

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ABSTRACT: In this paper, we show that time dilation is not a property of a single system but a comparison between two systems, defined solely by their relative spatial distribution of mass. Specifically stated, the time dilation between two systems is governed by the relative DeGerlia Inertial Density D , defined as mass divided by mean radius about the compared axis. We first exemplify this comparative nature with one of the most fundamental equivalences in General Relativity, the gravitational time-dilation equation, which computes a system's time dilation relative to flat space-time: $d\tau/dt = \sqrt{1 - r_s/r}$. We use this reference value to validate the calculation we perform using relative D alone. We then validate correct calculations for a diverse set of two-system examples, including transformed systems (similar to the gravitational time dilation above, where system 2 is a transformed version of system 1) as well as discrete system comparisons, across the classical and relativistic regimes. To conduct this demonstration, we introduce the DeGerlia Time Dilation Framework (DTDF), a simple, mathematically equivalent framework, fully defined by general relativity, that offers a second frame for visualizing the relative nature of time.

Keywords: time dilation, general relativity, special relativity, gravitational time dilation, Lorentz time dilation, DeGerlia inertial density, inertial density, moment of inertia, astrophysics, Schwarzschild radius, DTDF, DTD, degerlia time dilation

1 Introduction

General Relativity¹ is a model for describing our universe that is fundamentally comparative: no quantity has physical meaning without a comparator. The pace of time is no exception: the time dilation between two systems is a product of the systems' relative geometric mass distributions. Directly from the gravitational time dilation (GTD) formula, one can derive the underlying two-system logic, and ultimately, GTD is simply a comparison between a system's pace of time relative to the pace of time of a selected benchmark system, which in this case is a system of the same mass, but at its Schwarzschild radius. The formula requires only one mass and one radius, the second system being derived

directly from the first. While this is the familiar representation, it obscures the comparative nature of relativity and implies that one need not compare two systems to characterize time. That implication is incorrect: there is no privileged perspective from which to characterize time. In this paper, we evaluate the physical property responsible for relativistic time dilation and explore the underlying formulation that quantifies the relative time dilation between two systems.

The relative flow of time between two systems is governed solely by their relative DeGerlia inertial density D . We demonstrate this across a range of uniform spherical systems, from the classical to the relativistic regime, by calculating each system's DeGerlia time dilation from its relative D , converting each to its gravitational time dilation, and taking their ratio. In each case this matches the ratio obtained by computing each system's gravitational time dilation independently from the standard Schwarzschild form, to the precision permitted by G . Given two

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Table 1 System properties for each pair comparison case.

Property	Case 1 $m = m_s \sim r_s$	Case 2 $m = m_s \sim \frac{4}{3}r_s$	Case 3 $r = r$	Case 4 $\rho = \rho$	Case 5 Classical	Case 6 $l = l$	Case 7 General
m_1 (kg)	2.00000e+30	1.00000e+30	1.00000e+30	1.00000e+30	8.00000e+0	1.00000e+32	1.00000e+20
m_2 (kg)	2.00000e+30	1.00000e+30	1.00000e+24	1.00000e+27	1.00000e+0	1.00000e+30	1.00000e+30
r_1 (m)	2.97049e+3	1.98031e+3	1.00000e+8	1.00000e+8	2.00000e+0	1.00000e+7	2.00000e-7
r_2 (m)	7.00000e+8	1.00000e+8	1.00000e+8	1.00000e+7	1.00000e+0	1.00000e+8	1.00000e+7
ρ_1 (kg/m ³)	1.82161e+19	3.07406e+19	2.38732e+5	2.38732e+5	2.38732e-1	2.38732e+10	2.98416e+39
ρ_2 (kg/m ³)	1.39203e+3	2.38732e+5	2.38732e-1	2.38732e+5	2.38732e-1	2.38732e+5	2.38732e+8
I_1 (kg·m ²)	7.05907e+36	1.56865e+36	4.00000e+45	4.00000e+45	1.28000e+1	4.00000e+45	1.60000e+6
I_2 (kg·m ²)	3.92000e+47	4.00000e+45	4.00000e+39	4.00000e+40	4.00000e-1	4.00000e+45	4.00000e+43
D_1 (kg/m)	6.73289e+26	5.04972e+26	1.00000e+22	1.00000e+22	4.00000e+0	1.00000e+25	5.00000e+26
D_2 (kg/m)	2.85714e+21	1.00000e+22	1.00000e+16	1.00000e+20	1.00000e+0	1.00000e+22	1.00000e+23
r_{s1} (m)	2.97046e+3	1.48523e+3	1.48523e+3	1.48523e+3	1.18819e-26	1.48523e+5	1.48523e-7
r_{s2} (m)	2.97046e+3	1.48523e+3	1.48523e-3	1.48523e+0	1.48523e-27	1.48523e+3	1.48523e+3
$D_{\text{norm},1}$	9.99991e-1	7.50001e-1	1.48523e-5	1.48523e-5	5.94093e-27	1.48523e-2	7.42617e-1
$D_{\text{norm},2}$	4.24352e-6	1.48523e-5	1.48523e-11	1.48523e-7	1.48523e-27	1.48523e-5	1.48523e-4
k_s	9.99990e-1	7.50000e-1	1.48523e-5	1.48523e-5	5.94093e-27	1.48523e-2	7.42616e-1
k_d	9.99991e-1	7.50001e-1	1.48523e-5	1.48523e-5	5.94093e-27	1.48523e-2	7.42617e-1
GTD _{s1}	1-9.96838e-1	1-5.00000e-1	1-7.42619e-6	1-7.42619e-6	1-2.97046e-27	1-7.45394e-3	1-4.92670e-1
GTD _{s2}	1-2.12176e-6	1-7.42619e-6	1-7.42616e-12	1-7.42616e-8	1-7.42616e-28	1-7.42619e-6	1-7.42644e-5
GTD _{d1}	1-9.96948e-1	1-5.00001e-1	1-7.42619e-6	1-7.42619e-6	1-2.97047e-27	1-7.45395e-3	1-4.92670e-1
GTD _{d2}	1-2.12176e-6	1-7.42619e-6	1-7.42617e-12	1-7.42617e-8	1-7.42617e-28	1-7.42619e-6	1-7.42644e-5
DTD ₁	9.99995e-1	8.66026e-1	3.85387e-3	3.85387e-3	7.70774e-14	1.21870e-1	8.61752e-1
DTD ₂	2.05998e-3	3.85387e-3	3.85387e-6	3.85387e-4	3.85387e-14	3.85387e-3	1.21870e-2

Table 2 Derived ratios across cases.

Ratio	Case 1 $m = m_s \sim r_s$	Case 2 $m = m_s \sim \frac{4}{3}r_s$	Case 3 $r = r$	Case 4 $\rho = \rho$	Case 5 Classical	Case 6 $l = l$	Case 7 General
DTD ₁ /DTD ₂	4.85439e+2	2.24716e+2	1.00000e+3	1.00000e+1	2.00000e+0	3.16228e+1	7.07107e+1
$\sqrt{D_1/D_2}$	4.85439e+2	2.24716e+2	1.00000e+3	1.00000e+1	2.00000e+0	3.16228e+1	7.07107e+1
$\sqrt{(m_1 r_2)/(m_2 r_1)}$	4.85439e+2	2.24716e+2	1.00000e+3	1.00000e+1	2.00000e+0	3.16228e+1	7.07107e+1
$\sqrt{(I_1/I_2)(r_2/r_1)^3}$	4.85439e+2	2.24716e+2	1.00000e+3	1.00000e+1	2.00000e+0	3.16228e+1	7.07107e+1
$(I_1/I_2)^{1/5}(\rho_1/\rho_2)^{3/10}$	4.85439e+2	2.24716e+2	1.00000e+3	1.00000e+1	2.00000e+0	3.16228e+1	7.07107e+1
$(m_1 r_2)/(m_2 r_1)$	2.35651e+5	5.04972e+4	1.00000e+6	1.00000e+2	4.00000e+0	1.00000e+3	5.00000e+3
D_1/D_2	2.35651e+5	5.04972e+4	1.00000e+6	1.00000e+2	4.00000e+0	1.00000e+3	5.00000e+3
$(I_1/I_2)(r_2/r_1)^3$	2.35651e+5	5.04972e+4	1.00000e+6	1.00000e+2	4.00000e+0	1.00000e+3	5.00000e+3
GTD _{s1} /GTD _{s2}	1-9.96838e-1	1-4.99996e-1	1-7.42618e-6	1-7.35193e-6	1-2.22785e-27	1-7.44657e-3	1-4.92632e-1
GTD _{d1} /GTD _{d2}	1-9.96948e-1	1-4.99997e-1	1-7.42619e-6	1-7.35193e-6	1-2.22785e-27	1-7.44658e-3	1-4.92633e-1
$\sqrt{1 - \text{DTD}_1^2}/\sqrt{1 - \text{DTD}_2^2}$	1-9.96948e-1	1-4.99997e-1	1-7.42619e-6	1-7.35193e-6	1-2.22785e-27	1-7.44658e-3	1-4.92633e-1
I_1/I_2	1.80078e-11	3.92163e-10	1.00000e+6	1.00000e+5	3.20000e+1	1.00000e+0	4.00000e-38
$(I_1/I_2)^{1/2}$	4.24356e-6	1.98031e-5	1.00000e+3	3.16228e+2	5.65685e+0	1.00000e+0	2.00000e-19
$(\rho_1/\rho_2)^{1/5}(r_1/r_2)$	7.09728e-3	1.31430e-2	1.58489e+1	1.00000e+1	2.00000e+0	1.00000e+0	3.31445e-8
$(I_1/I_2)^{1/5}$	7.09728e-3	1.31430e-2	1.58489e+1	1.00000e+1	2.00000e+0	1.00000e+0	3.31445e-8
m_1/m_2	1.00000e+0	1.00000e+0	1.00000e+6	1.00000e+3	8.00000e+0	1.00000e+2	1.00000e-10
$\sqrt{m_1/m_2}$	1.00000e+0	1.00000e+0	1.00000e+3	3.16228e+1	2.82843e+0	1.00000e+1	1.00000e-5
r_1/r_2	4.24356e-6	1.98031e-5	1.00000e+0	1.00000e+1	2.00000e+0	1.00000e-1	2.00000e-14
$\sqrt{r_1/r_2}$	2.05999e-3	4.45007e-3	1.00000e+0	3.16228e+0	1.41421e+0	3.16228e-1	1.41421e-7

systems' masses and radii, their relative time dilation is determined by their relative spatial distribution of mass alone, a relation already contained in general relativity.

2 Experimental Model

The experimental model employed in this study comprises several components. The framework is derived directly from general relativity and is mathematically equivalent to established general-relativistic calculations. It neither relies upon nor introduces any principle not already established within the theories of general and special relativity; this paper leverages a framing of the existing theory of relativity for illustrative purposes.

We begin by introducing the DeGerlia Framework and then demonstrate its complete mathematical equivalence with general relativity. We then work through a series of examples, each demonstrating a different comparative case, and in each case, we validate the result using standard gravitational time-dilation calculations.

These examples demonstrate the two-system nature of all relativistic calculations. Relativity is fundamentally concerned with how quantities appear relative to one another; gravitational time dilation, for instance, reflects the comparison of a system with an identical system at its Schwarzschild radius, the collapse-condition radius r_s .

3 The DeGerlia Time Dilation Framework

3.1 Definitions, Formulas and Constants Used Herein

Symbol	Definition / Value
c	Light Speed $299\,792\,458\text{ m s}^{-1}$
D	DeGerlia Inertial Density m/r , where r is the mean radius about a selected axis
D_{crit}	DeGerlia Threshold $c^2/2G = 6.73295\text{e}26\text{ kg/m}$
DTD	DeGerlia Time Dilation $\sqrt{k}$
G	Gravitational Constant $6.674\,30\text{e}-11 \pm 1.5\text{e}-15\text{ m}^3\text{ kg}^{-1}\text{ s}^{-2}$
GTD	Gravitational Time Dilation $d\tau/dt = \sqrt{1-k}$
k	Linear Scale Factor $v^2/c^2 = r_s/r = D/D_{\text{crit}}$
k_D	Linear Scale Factor (DeGerlia inertial density) D/D_{crit}
k_s	Linear Scale Factor (Schwarzschild) r_s/r
k_v	Linear Scale Factor (velocity) v^2/c^2
P	Inertial Density $I/V = D \cdot 3/10\pi$
r_s	Schwarzschild Radius $2Gm/c^2$

3.2 What is the DeGerlia Time Dilation Framework?

First and foremost, the entirety of the principles described herein represents no deviation from general relativity. It represents a mathematically equivalent reframing useful in certain relativistic calculations. The author has simply asserted a complement to GTD, where the form of time dilation is equal to $\sqrt{k}$, instead of $\sqrt{1-k}$.

While equivalent to GTD mathematically, the DTD form offers a different perspective on the same physical phenomenon. It is a complementary perspective that aids in visualizing the underlying

physics:

- One can convert between GTD and DTD at any time. GTD is always recoverable.
- DTD is mathematically very easy to use and visualize, requiring only the ratio of mass to radius in comparison to a static constant to perform gravitational time dilation calculations.
- DTD allows for direct calculation of time dilation between two systems, without the need to reference a third system (the Schwarzschild condition)
- DTD returns finite output over the entire range of finite input.
- DTD represents flat spacetime as an ideal state that can only be approached.
- DTD represents the Schwarzschild condition as the point where the gravitational escape velocity exceeds the speed of light, and an event horizon forms. While we have observed black holes, the asymptotically approached ideal state of zero time dilation is not attainable.
- DTD reflects the singularity state as an asymptotic approach to $r = 0$.

3.3 Inertia and Moment of Inertia

In classical physics, inertia dictates how a system's motion changes in response to an applied force. This applies to both linear and rotational motion. In rotational dynamics, the "moment of inertia"² I dictates how the angular motion of a system about a specific axis changes when a torque is applied, and is defined as the sum of each mass element multiplied by the square of its distance from the axis of rotation.

$$\text{Moment of Inertia: } I = \sum_i m_i r_i^2 \quad (1)$$

Using Equation (1), one can determine how a system's motion changes when energy is applied about an axis (torque). As it pertains to linear motion, inertia (I) is characterized solely by the system's mass.

Inertia and moment of inertia have several notable properties.

1. Inertia is dimensional: it exists in, and has a component within, each of the three spatial dimensions.
2. Inertia is multimodal: it may be linear or rotational. Linear inertia reflects the mass alone, whereas rotational inertia reflects the mass times the square of its radius about an axis.
3. The moment of inertia is additive about a given axis: if a system is divided into parts and each part's moment of inertia is measured about that same axis, the total is unchanged.
4. Inertia relates force to motion: it determines the change in motion produced by an applied force or torque, making it the bridge between the two. Lower inertia yields greater motion for a given force.

3.4 General Relativity and Time Dilation

General Relativity provides the formula for calculating a system's time dilation relative to "flat space-time".

$$d\tau/dt = \sqrt{1-k} \quad (2)$$

where $k = r_s/r = 2Gm/(rc^2)$, m is the system's mass, and r its radius.

The equation requires only a single system's mass and radius, yet k is itself a ratio of two radii: the system's radius r and the Schwarzschild radius r_s (11) of that same mass. The calculation is therefore a comparison of two systems: the system as given, and the same mass at its Schwarzschild radius.

3.5 Definition of a System

A "system" is defined as any discrete collection of particles and the space between them, or equivalently, any unit of space and the particles contained within it. The framework regards the universe not as a clean separation of discrete material systems separated by the vacuum of space, but as a gradient of material density pervasive throughout. The most extreme densities reside not within near-scale systems but within the cores of highly dense gravitational objects far from the human scale, such as black holes or subatomic particles, and form a gradient extending to the near-scale "vacuum of space" at which another gravitational source begins to dominate. The framework imposes no requirement for uniformity or symmetry, a clean boundary separating matter from vacuum, or any particular density profile. It requires only a mass and a radius about a chosen axis. A system may be specified, for example, as the Earth out to the mean radius of its surface. For a body such as Jupiter, which lacks a clear boundary separating surface from atmosphere, the system may instead be specified out to a chosen radius or density. For a large system, a spherical sample may be taken from within it to characterize the behavior of particular regions. There can be no system modeled under General Relativity that lacks either mass or radius.

3.6 System Time

For a whole system (as distinct from any subsystem), and from that whole system's perspective, the pace of time is static and all other systems are perceived relative to that benchmark. A subsystem possesses distinct physical properties and consequently exhibits behavior different from that of the system as a whole. In general relativity, time dilation scales precisely with the relative mass-to-radius ratio; a part of a system, therefore, cannot be expected to share the pace of time of the whole.

3.7 Inertial Density P

Inertial density P as shown in (4) is a fundamental system property³ characterizing the distribution of inertia across any system, whether in the linear or rotational context, in any direction, or about any axis. If we were to redefine classical density as I/V (inertia/volume), because inertia is mass for one-dimensional motion, this would remain consistent with the classical definition of density, where $\rho = m/V$.

Inertial density characterizes how a system's mass is distributed relative to its axis, for a given mass and volume. Consider the configuration with the greatest moment of inertia for a given volume: a thin shell, in which all mass resides at the radius, displaced as far from the axis as the geometry permits. A soap bubble exemplifies this configuration; per unit volume it possesses substantial inertia despite its low mass, representing the maximum-inertia structure for its volume. Conversely, a configuration of equal radius with its mass concentrated as an approximate point mass at the center represents the opposite limit: as the mass concentrates toward the center, the moment of inertia approaches zero. The uniform solid sphere lies between these limits. What determines the inertia about a given axis is the mean radius of the mass distribution about that axis; the internal density gradient that produces a given mean radius does not enter, just as ordinary density averages over a system's internal structure and the density of any subset is obtained by evaluating that subset directly.

3.8 Mathematics of Inertial Density

To derive the inertial density relationship, we begin with the established relationship between mass and density ρ as shown in (3):

$$\rho = \frac{m}{V} \quad (3)$$

where ρ is density, m is mass, and V is volume. Because mass is equivalent to linear inertia, and because density is m/V , we define inertial density P in terms of moment of inertia over volume:

$$\text{Inertial Density : } P = \frac{I}{V} \quad (4)$$

where inertial density P is equal to inertia I over volume V . Note, the moment of inertia I for a system may be distinct about every axis for systems that are not uniform⁴. For a uniform sphere⁵, we substitute $I = \frac{2}{5}mr^2$ and $V = \frac{4}{3}\pi r^3$:

$$P = \frac{I}{V} = \frac{\frac{2}{5}mr^2}{\frac{4}{3}\pi r^3} = \frac{3m}{10\pi r} \quad (5)$$

Which can be simplified as follows:

$$P = \frac{m}{r} \times \frac{3}{10\pi} \quad (6)$$

$$P = \frac{m}{r} \times 9.549296585e-2 \quad (7)$$

Therefore, for spherical systems, P can be calculated as the ratio of mass to radius, multiplied by the conversion factor $\frac{3}{10\pi}$ as shown in (7).

3.9 DeGerlia Inertial Density D

For all spherical systems, m/r multiplied by the geometric factor $3/(10\pi)$ yields the inertial density. *DeGerlia inertial density* is a simplified analog of inertial density I/V , equal to the mass-to-radius ratio m/r with the $3/(10\pi)$ geometric conversion factor

omitted:

$$\text{DeGerlia Inertial Density: } D = \frac{m}{r} \quad (8)$$

DeGerlia inertial density (8) $D = m/r$ is related to inertial density P by the geometric factor $3/10\pi$ as shown in (9):

$$P = \frac{3}{10\pi} \times D \quad (9)$$

In a linear context, inertia is mass, and inertial density reduces directly to mass density $\rho = m/V$. From this perspective, inertial density is the rotational analog of mass density, paralleling the relationship between $F = ma$ and $\tau = I\alpha$. Equivalently, mass density is the linear interpretation of inertial density.

Just as ordinary density ($\rho = m/V$) is linear inertia (mass) distributed over volume, DeGerlia inertial density is rotational inertia distributed over volume. As stated in the prior work, *mass and density are to linear motion as moment of inertia and inertial density are to rotational motion*.

Because D is almost always used in a ratio that cancels all geometric constants, this metric is typically preferred over the geometrically accurate inertial density P for ease of use.

3.10 DeGerlia Threshold D_{crit}

The Schwarzschild condition can be expressed as a static universal constant threshold of DeGerlia inertial density D , rather than as a value that must be recalculated for every mass.

Setting $r = r_s$ (the Schwarzschild condition), the DeGerlia inertial density D becomes the threshold D_{crit} :

$$D_{\text{crit}} = \frac{m}{r_s} \quad (10)$$

Recall the Schwarzschild radius⁶:

$$r_s = \frac{2Gm}{c^2} \quad (11)$$

Substitute r_s into the expression for D_{crit} :

$$D_{\text{crit}} = \frac{m}{\frac{2Gm}{c^2}} = \frac{mc^2}{2Gm} = \frac{c^2}{2G} \quad (12)$$

Insert the fundamental constants⁷, $c = 2.99792458\text{e}8\text{ m s}^{-1}$ and $G = 6.67430\text{e}-11\text{ m}^3\text{ kg}^{-1}\text{ s}^{-2}$:

$$\begin{aligned} D_{\text{crit}} &= \frac{(2.99792458\text{e}8)^2}{2 \times 6.67430\text{e}-11} \text{ kg/m} \\ &= \frac{8.98755179\text{e}16}{1.334860\text{e}-10} \\ &= 6.73295\text{e}26 \text{ kg/m} \end{aligned}$$

$$\text{DeGerlia Threshold: } D_{\text{crit}} = 6.73295\text{e}26 \text{ kg/m} \quad (13)$$

The DeGerlia Threshold (13) is the universal constant value of DeGerlia inertial density D at which the escape velocity equals the speed of light: the mass-over-radius threshold at which gravitational collapse (collapse of the neutron structure) occurs.

Note this value is limited by the precision of the gravitational constant⁸, $G = 6.67430\text{e}-11 \pm 1.5\text{e}-15\text{ m}^3\text{ kg}^{-1}\text{ s}^{-2}$. Just as the Schwarzschild radius, the DeGerlia threshold is universally applicable to spherical gravitational systems; examples include black holes or neutron stars, a spherical region of an interstellar cloud, or even a galactic cluster.

Rapid evaluation of the black hole condition. For a single system, divide its mass by its radius and compare the result with the DeGerlia Threshold (13), $D_{\text{crit}} = 6.73295\text{e}26 \text{ kg/m}$. If their ratio equals 1, the system is at the black hole condition; if it is less than 1, no collapse has occurred.

3.11 DeGerlia Time Dilation

DeGerlia Time Dilation (DTD) (14) is the Pythagorean complement (15) of gravitational time dilation:

$$\text{GTD} = \sqrt{1-k}$$

$$\text{DTD} = \sqrt{k} \quad (14)$$

$$\text{GTD}^2 + \text{DTD}^2 = 1 \quad (15)$$

where k is the linear scale factor.

3.12 The Linear Scale Factor k

The linear scale factor k is a common element among all established forms of time dilation. Time dilation is always:

$$\frac{d\tau}{dt} = \sqrt{1-k}$$

where $d\tau/dt$ is the time-dilation factor between the two systems and k is the linear scale factor between the two systems. Specific forms of k follow:

- $k = r_s/r$, the standard Schwarzschild condition.
- $k = D/D_{\text{crit}} = (m/r)/(6.73295\text{e}26 \text{ kg/m})$, via the DeGerlia inertial density against its threshold.
- $k = v^2/c^2$, as used in special relativity.

4 Relative Time Dilation Examples

Tables 1 and 2 present the complete system properties and derived ratios for seven system pairs spanning the cases that follow: equal mass at the Schwarzschild radius, equal mass at four-thirds the Schwarzschild radius, equal radius, equal density, a classical case, equal moment of inertia, and a general unconstrained case.

Across every case, the time-dilation ratio computed via the DeGerlia inertial density ratio agrees with the gravitational time-dilation ratio computed independently from the standard Schwarzschild form, to the limit of precision imposed by G . No case requires a separate formula. The worked examples in the following subsections show the full calculations, case by case.

4.1 Case 1: ($m_1 = m_2$, $r_1 \approx r_s$)

The two identical masses cancel, reducing the DeGerlia inertial density ratio to a ratio of radii. Note that this reflects the k calculation for gravitational time dilation, where we compare with r_s . Because $GTD = 0$ exactly at r_s , we set the comparison radius to $1.00001 r_s$, an "arbitrarily" small amount above r_s .

System properties:

	System 1 (S_1)	System 2 (S_2)
m (kg)	2.00000e30	2.00000e30
r (m)	2.97049e3	7.00000e8

Calculate DTD for each System:

$$\begin{aligned}
 DTD &= \sqrt{k} \quad \text{where} \quad k = m/(rD_{\text{crit}}) \\
 k_1 &= \frac{2.00000e30 \text{ kg}}{(2.97049e3 \text{ m})(6.73295e26 \text{ kg/m})} = 9.9999068441e-1 \\
 DTD_1 &= \sqrt{9.9999068441e-1} = 9.9999534220e-1 \\
 k_2 &= \frac{2.00000e30 \text{ kg}}{(7.00000e8 \text{ m})(6.73295e26 \text{ kg/m})} = 4.24352e-6 \\
 DTD_2 &= \sqrt{4.24352e-6} = 2.05998e-3 \\
 \frac{DTD_1}{DTD_2} &= \frac{9.9999534220e-1}{2.05998e-3} = 4.85439e2 \\
 \frac{DTD_2}{DTD_1} &= \frac{2.05998e-3}{9.9999534220e-1} = 2.05998e-3
 \end{aligned}$$

Calculate the GTD ratio from DTD:

$$\begin{aligned}
 \frac{GTD_{d1}}{GTD_{d2}} &= \frac{\sqrt{1 - DTD_1^2}}{\sqrt{1 - DTD_2^2}} = \frac{\sqrt{1 - (9.9999534220e-1)^2}}{\sqrt{1 - (2.05998e-3)^2}} \\
 &= \frac{\sqrt{1 - 9.9999068441e-1}}{\sqrt{1 - 4.24352e-6}} \\
 &= \frac{\sqrt{9.31559e-6}}{\sqrt{9.9999575648e-1}} \\
 &= \frac{3.05214e-3}{9.9999787824e-1} = 3.05215e-3 \\
 &= 1 - 9.9694785e-1
 \end{aligned}$$

Verify by calculating the GTD from the Schwarzschild Radius Ratio:
Schwarzschild radii:

$$\begin{aligned}
 r_{s1} = r_{s2} &= \frac{2Gm}{c^2} = \frac{2(6.67430e-11 \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2})(2.00000e30 \text{ kg})}{(2.99792e8 \text{ m s}^{-1})^2} \\
 &= 2.97046e3 \text{ m}
 \end{aligned}$$

Calculate k values from the r_s/r ratio:

$$\begin{aligned}
 k_1 &= \frac{r_{s1}}{r_1} = \frac{2.97046e3 \text{ m}}{2.97049e3 \text{ m}} = 9.999900010e-1 \\
 k_2 &= \frac{r_{s2}}{r_2} = \frac{2.97046e3 \text{ m}}{7.00000e8 \text{ m}} = 4.24352e-6
 \end{aligned}$$

Calculate GTD via Schwarzschild radius ratio:

$$\begin{aligned}
 GTD_{s1} &= \sqrt{1 - k_1} = \sqrt{9.99990e-6} = 3.16226e-3 = 1 - 9.9683774e-1 \\
 GTD_{s2} &= \sqrt{1 - k_2} = \sqrt{9.9999575648e-1} = 9.9999787824e-1 \\
 &= 1 - 2.12176e-6 \\
 \frac{GTD_{s1}}{GTD_{s2}} &= \frac{3.16226e-3}{9.9999787824e-1} = 3.16227e-3 \\
 \frac{GTD_{s2}}{GTD_{s1}} &= \frac{9.9999787824e-1}{3.16226e-3} = 3.16227e-3
 \end{aligned}$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\begin{aligned}
 \frac{GTD_{s1}}{GTD_{s2}} &= 1 - 9.9683773e-1 \quad (\text{via } r_s) \\
 \frac{GTD_{d1}}{GTD_{d2}} &= 1 - 9.9694785e-1 \quad (\text{via } D_{\text{crit}})
 \end{aligned}$$

4.2 Case 2: ($m_1 = m_2$, $r_1 = \frac{4}{3}r_s$)

The two identical masses cancel, reducing the DeGerlia inertial density ratio to a ratio of radii. Note: This reflects the k calculation for gravitational time dilation. However, in this case, instead of comparing with r_s , we compare with $\frac{4}{3}r_s$, which corresponds to a gravitational time dilation of $GTD = 5.00000e-1$.

System properties:

	System 1 (S_1)	System 2 (S_2)
m (kg)	1.00000e30	1.00000e30
r (m)	1.98031e3	1.00000e8

Calculate DTD for each System:

$$\begin{aligned}
 DTD &= \sqrt{k} \quad \text{where} \quad k = m/(rD_{\text{crit}}) \\
 k_1 &= \frac{1.00000e30 \text{ kg}}{(1.98031e3 \text{ m})(6.73295e26 \text{ kg/m})} = 7.50001e-1 \\
 DTD_1 &= \sqrt{7.50001e-1} = 8.66026e-1 \\
 k_2 &= \frac{1.00000e30 \text{ kg}}{(1.00000e8 \text{ m})(6.73295e26 \text{ kg/m})} = 1.48523e-5 \\
 DTD_2 &= \sqrt{1.48523e-5} = 3.85387e-3 \\
 \frac{DTD_1}{DTD_2} &= \frac{8.66026e-1}{3.85387e-3} = 2.24716e2 \\
 \frac{DTD_2}{DTD_1} &= \frac{3.85387e-3}{8.66026e-1} = 4.45000e-3
 \end{aligned}$$

Calculate the GTD ratio from DTD:

$$\begin{aligned}
 \frac{GTD_{d1}}{GTD_{d2}} &= \frac{\sqrt{1 - DTD_1^2}}{\sqrt{1 - DTD_2^2}} = \frac{\sqrt{1 - (8.66026e-1)^2}}{\sqrt{1 - (3.85387e-3)^2}} \\
 &= \frac{\sqrt{1 - 7.50001e-1}}{\sqrt{1 - 1.48523e-5}} \\
 &= \frac{\sqrt{2.49999e-1}}{\sqrt{9.9999851477e-1}} \\
 &= \frac{4.99999e-1}{9.9999257381e-1} = 5.00003e-1 \\
 &= 1 - 4.99997e-1
 \end{aligned}$$

Verify by calculating the GTD from the Schwarzschild Radius Ratio:
Schwarzschild radii:

$$\begin{aligned}
 r_{s1} = r_{s2} &= \frac{2Gm}{c^2} = \frac{2(6.67430e-11 \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2})(1.00000e30 \text{ kg})}{(2.99792e8 \text{ m s}^{-1})^2} \\
 &= 1.48523e3 \text{ m}
 \end{aligned}$$

Calculate k values from the r_s/r ratio:

$$\begin{aligned}
 k_1 &= \frac{r_{s1}}{r_1} = \frac{1.48523e3 \text{ m}}{1.98031e3 \text{ m}} = 7.50000e-1 \\
 k_2 &= \frac{r_{s2}}{r_2} = \frac{1.48523e3 \text{ m}}{1.00000e8 \text{ m}} = 1.48523e-5
 \end{aligned}$$

Calculate GTD via Schwarzschild radius ratio:

$$\begin{aligned}
 GTD_{s1} &= \sqrt{1 - k_1} = \sqrt{2.50000e-1} = 5.00000e-1 = 1 - 5.00000e-1 \\
 GTD_{s2} &= \sqrt{1 - k_2} = \sqrt{9.9999851477e-1} = 9.9999257381e-1 \\
 &= 1 - 7.42619e-6 \\
 \frac{GTD_{s1}}{GTD_{s2}} &= \frac{5.00000e-1}{9.9999257381e-1} = 5.00004e-1 \\
 \frac{GTD_{s2}}{GTD_{s1}} &= \frac{9.9999257381e-1}{5.00000e-1} = 1 - 4.99996e-1
 \end{aligned}$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\begin{aligned}
 \frac{GTD_{s1}}{GTD_{s2}} &= 1 - 4.99996e-1 \quad (\text{via } r_s) \\
 \frac{GTD_{d1}}{GTD_{d2}} &= 1 - 4.99997e-1 \quad (\text{via } D_{\text{crit}})
 \end{aligned}$$

4.3 Case 3: ($r_1 = r_2$)

When the two systems share the same radius, the radius terms cancel. The Lorentz factor⁹ has the same $\sqrt{1-k}$ form, with $k = v^2/c^2$. Since velocity is distance per unit time, v^2/c^2 is the square of a ratio of two lengths, with the shared time unit canceling.

System properties:

	System 1 (S_1)	System 2 (S_2)
m (kg)	1.00000e30	1.00000e24
r (m)	1.00000e8	1.00000e8

Calculate DTD for each System:

$$\begin{aligned}
 DTD &= \sqrt{k} \quad \text{where} \quad k = m/(rD_{\text{crit}}) \\
 k_1 &= \frac{1.00000e30 \text{ kg}}{(1.00000e8 \text{ m})(6.73295e26 \text{ kg/m})} = 1.48523e-5 \\
 DTD_1 &= \sqrt{1.48523e-5} = 3.85387e-3 \\
 k_2 &= \frac{1.00000e24 \text{ kg}}{(1.00000e8 \text{ m})(6.73295e26 \text{ kg/m})} = 1.48523e-11 \\
 DTD_2 &= \sqrt{1.48523e-11} = 3.85387e-6 \\
 \frac{DTD_1}{DTD_2} &= \frac{3.85387e-3}{3.85387e-6} = 1.00000e3 \\
 \frac{DTD_2}{DTD_1} &= \frac{3.85387e-6}{3.85387e-3} = 1.00000e-3
 \end{aligned}$$

Calculate the GTD ratio from DTD:

$$\begin{aligned}
 \frac{GTD_{d1}}{GTD_{d2}} &= \frac{\sqrt{1-DTD_1^2}}{\sqrt{1-DTD_2^2}} = \frac{\sqrt{1-(3.85387e-3)^2}}{\sqrt{1-(3.85387e-6)^2}} \\
 &= \frac{\sqrt{1-1.48523e-5}}{\sqrt{1-1.48523e-11}} \\
 &= \frac{\sqrt{9.999851477e-1}}{\sqrt{9.999999999851477e-1}} \\
 &= \frac{9.9999257381e-1}{9.999999999257381e-1} = 9.9999257381e-1 \\
 &= 1 - 7.42619e-6
 \end{aligned}$$

Verify by calculating the GTD from the Schwarzschild Radius Ratio: Schwarzschild radii:

$$\begin{aligned}
 r_{s1} &= \frac{2Gm_1}{c^2} = \frac{2(6.67430e-11 \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2})(1.00000e30 \text{ kg})}{(2.99792e8 \text{ m s}^{-1})^2} \\
 &= 1.48523e3 \text{ m} \\
 r_{s2} &= \frac{2Gm_2}{c^2} = \frac{2(6.67430e-11 \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2})(1.00000e24 \text{ kg})}{(2.99792e8 \text{ m s}^{-1})^2} \\
 &= 1.48523e-3 \text{ m}
 \end{aligned}$$

Calculate k values from the r_s/r ratio:

$$\begin{aligned}
 k_1 &= \frac{r_{s1}}{r_1} = \frac{1.48523e3 \text{ m}}{1.00000e8 \text{ m}} = 1.48523e-5 \\
 k_2 &= \frac{r_{s2}}{r_2} = \frac{1.48523e-3 \text{ m}}{1.00000e8 \text{ m}} = 1.48523e-11
 \end{aligned}$$

Calculate GTD via Schwarzschild radius ratio:

$$\begin{aligned}
 GTD_{s1} &= \sqrt{1-k_1} = \sqrt{9.999851477e-1} = 9.9999257381e-1 \\
 &= 1 - 7.42619e-6 \\
 GTD_{s2} &= \sqrt{1-k_2} = \sqrt{9.999999999851477e-1} \\
 &= 9.999999999257384e-1 = 1 - 7.42616e-12 \\
 \frac{GTD_{s1}}{GTD_{s2}} &= \frac{9.9999257381e-1}{9.999999999257384e-1} \\
 &= 9.9999257382e-1 = 1 - 7.42618e-6
 \end{aligned}$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\begin{aligned}
 \frac{GTD_{s1}}{GTD_{s2}} &= 1 - 7.42618e-6 \quad (\text{via } r_s) \\
 \frac{GTD_{d1}}{GTD_{d2}} &= 1 - 7.42619e-6 \quad (\text{via } D_{\text{crit}})
 \end{aligned}$$

4.4 Case 4: ($\rho_1 = \rho_2$)

When the two systems share the same density, $m \propto r^3$, so $m_1/m_2 = (r_1/r_2)^3$.

System properties:

	System 1 (S_1)	System 2 (S_2)
m (kg)	1.00000e30	1.00000e27
r (m)	1.00000e8	1.00000e7

Calculate DTD for each System:

$$\begin{aligned}
 DTD &= \sqrt{k} \quad \text{where} \quad k = m/(rD_{\text{crit}}) \\
 k_1 &= \frac{1.00000e30 \text{ kg}}{(1.00000e8 \text{ m})(6.73295e26 \text{ kg/m})} = 1.48523e-5 \\
 DTD_1 &= \sqrt{1.48523e-5} = 3.85387e-3 \\
 k_2 &= \frac{1.00000e27 \text{ kg}}{(1.00000e7 \text{ m})(6.73295e26 \text{ kg/m})} = 1.48523e-7 \\
 DTD_2 &= \sqrt{1.48523e-7} = 3.85387e-4 \\
 \frac{DTD_1}{DTD_2} &= \frac{3.85387e-3}{3.85387e-4} = 1.00000e1 \\
 \frac{DTD_2}{DTD_1} &= \frac{3.85387e-4}{3.85387e-3} = 1.00000e-1
 \end{aligned}$$

Calculate the GTD ratio from DTD:

$$\begin{aligned}
 \frac{GTD_{d1}}{GTD_{d2}} &= \frac{\sqrt{1-DTD_1^2}}{\sqrt{1-DTD_2^2}} = \frac{\sqrt{1-(3.85387e-3)^2}}{\sqrt{1-(3.85387e-4)^2}} \\
 &= \frac{\sqrt{1-1.48523e-5}}{\sqrt{1-1.48523e-7}} \\
 &= \frac{\sqrt{9.999851477e-1}}{\sqrt{9.99999851477e-1}} \\
 &= \frac{9.9999257381e-1}{9.99999257383e-1} = 9.9999264807e-1 \\
 &= 1 - 7.35193e-6
 \end{aligned}$$

Verify by calculating the GTD from the Schwarzschild Radius Ratio: Schwarzschild radii:

$$\begin{aligned}
 r_{s1} &= \frac{2Gm_1}{c^2} = \frac{2(6.67430e-11 \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2})(1.00000e30 \text{ kg})}{(2.99792e8 \text{ m s}^{-1})^2} \\
 &= 1.48523e3 \text{ m} \\
 r_{s2} &= \frac{2Gm_2}{c^2} = \frac{2(6.67430e-11 \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2})(1.00000e27 \text{ kg})}{(2.99792e8 \text{ m s}^{-1})^2} \\
 &= 1.48523 \text{ m}
 \end{aligned}$$

Calculate k values from the r_s/r ratio:

$$\begin{aligned}
 k_1 &= \frac{r_{s1}}{r_1} = \frac{1.48523e3 \text{ m}}{1.00000e8 \text{ m}} = 1.48523e-5 \\
 k_2 &= \frac{r_{s2}}{r_2} = \frac{1.48523 \text{ m}}{1.00000e7 \text{ m}} = 1.48523e-7
 \end{aligned}$$

Calculate GTD via Schwarzschild radius ratio:

$$\begin{aligned}
 GTD_{s1} &= \sqrt{1-k_1} = \sqrt{9.999851477e-1} = 9.9999257381e-1 \\
 &= 1 - 7.42619e-6 \\
 GTD_{s2} &= \sqrt{1-k_2} = \sqrt{9.99999851477e-1} = 9.99999257384e-1 \\
 &= 1 - 7.42616e-8 \\
 \frac{GTD_{s1}}{GTD_{s2}} &= \frac{9.9999257381e-1}{9.99999257384e-1} \\
 &= 9.9999264807e-1 = 1 - 7.35193e-6
 \end{aligned}$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\begin{aligned}
 \frac{GTD_{s1}}{GTD_{s2}} &= 1 - 7.35193e-6 \quad (\text{via } r_s) \\
 \frac{GTD_{d1}}{GTD_{d2}} &= 1 - 7.35193e-6 \quad (\text{via } D_{\text{crit}})
 \end{aligned}$$

4.5 Case 5: (Classical)

This is a static-density, isometric case that follows the same model as Case 4.

System properties:

	System 1 (S_1)	System 2 (S_2)
m (kg)	8.00000	1.00000
r (m)	2.00000	1.00000

Calculate DTD for each System:

$$DTD = \sqrt{k} \quad \text{where} \quad k = m/(rD_{\text{crit}})$$

$$k_1 = \frac{8.00000 \text{ kg}}{(2.00000 \text{ m})(6.73295 \text{ e}26 \text{ kg/m})} = 5.94093 \text{ e} -27$$

$$DTD_1 = \sqrt{5.94093e-27} = 7.70774e-14$$

$$k_2 = \frac{1.00000\text{kg}}{(1.00000\text{m})(6.73295\text{e}26\text{kg/m})} = 1.48523\text{e-}27$$

$$DTD_2 = \sqrt{1.48523e-27} = 3.85387e-14$$

$$\frac{DTD_1}{\text{}} = \frac{7.70774\text{e-}14}{\text{}} = 2.00000$$

$$\overline{DTD_2} = \overline{3.85387e-14} = 2.00000$$

Calculate the GTD ratio from DTD:

[illegible]

Verify by calculating the GTD from the Schwarzschild Radius Ratio:

Schwarzschild radii:

$$\begin{aligned} r_{s1} &= \frac{2Gm_1}{c^2} = \frac{2(6.67430\text{e}-11\text{ m}^3\text{ kg}^{-1}\text{ s}^{-2})(8.00000\text{ kg})}{(2.99792\text{e}8\text{ m s}^{-1})^2} \\ &= 1.18819\text{e}-26\text{ m} \\ r_{s2} &= \frac{2Gm_2}{c^2} = \frac{2(6.67430\text{e}-11\text{ m}^3\text{ kg}^{-1}\text{ s}^{-2})(1.00000\text{ kg})}{(2.99792\text{e}8\text{ m s}^{-1})^2} \\ &= 1.48523\text{e}-27\text{ m} \end{aligned}$$

Calculate k values from the r_s/r ratio:

$$\begin{aligned} k_1 &= \frac{r_{s1}}{r_1} = \frac{1.18819 \text{e-}26 \text{m}}{2.00000 \text{m}} = 5.94093 \text{e-}27 \\ k_2 &= \frac{r_{s2}}{r_2} = \frac{1.48523 \text{e-}27 \text{m}}{1.00000 \text{m}} = 1.48523 \text{e-}27 \end{aligned}$$

Calculate GTD via Schwarzschild radius ratio:

[illegible]

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\frac{GTD_{s1}}{GTD_{s2}} = 1 - 2.22785\text{e-}27 \quad (\text{via } r_s)$$

$$\frac{GTD_{d1}}{GTD_{d2}} = 1 - 2.22785\text{e-}27 \quad (\text{via } D_{\text{crit}})$$

4.6 Case 6: ($I_1 = I_2$)

When the two systems share the same moment of inertia, $m_1 r_1^2 = m_2 r_2^2$, so $m_1/m_2 = (r_2/r_1)^2$. The DeGerlia inertial density ratio reduces to the cube of the radius ratio.

System properties:

	System 1 (S_1)	System 2 (S_2)
m (kg)	1.00000e32	1.00000e30
r (m)	1.00000e7	1.00000e8

Calculate DTD for each System:

$$DTD = \sqrt{k} \quad \text{where} \quad k = m/(rD_{\text{crit}})$$

$$k_1 = \frac{1.00000\text{e}32\text{ kg}}{(1.00000\text{e}7\text{ m})(6.73295\text{e}26\text{ kg/m})} = 1.48523\text{e}-2$$

$$DTD_1 = \sqrt{1.48523e-2} = 1.21870e-1$$

$$k_2 = \frac{1.00000\text{e}30\text{kg}}{(1.00000\text{e}8\text{m})(6.73295\text{e}26\text{kg/m})} = 1.48523\text{e}-5$$

$$DTD_2 = \sqrt{1.48523e-5} = 3.85387e-3$$

$DTD_1 = 1.21870e-1 \quad -3.16228e1$

$$\overline{DTD_2} = \overline{3.85387e-3} = 3.16228e1$$

Calculate the GTD ratio from DTD:

$$\begin{aligned} \frac{GTD_{d1}}{GTD_{d2}} &= \frac{\sqrt{1-DTD_1^2}}{\sqrt{1-DTD_2^2}} = \frac{\sqrt{1-(1.21870e-1)^2}}{\sqrt{1-(3.85387e-3)^2}} \\ &= \frac{\sqrt{1-1.48523e-2}}{\sqrt{1-1.48523e-5}} \\ &= \frac{\sqrt{9.851477e-1}}{\sqrt{9.999851477e-1}} \\ &= \frac{9.9254605e-1}{9.999925738e-1} = 9.9255342e-1 \\ &= 1-7.44658e-3 \end{aligned}$$

Verify by calculating the GTD from the Schwarzschild Radius Ratio:

Schwarzschild radii:

$$\begin{aligned} r_{s1} &= \frac{2Gm_1}{c^2} = \frac{2(6.67430\text{e}-11 \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2})(1.00000\text{e}32 \text{ kg})}{(2.99792\text{e}8 \text{ m s}^{-1})^2} \\ &= 1.48523\text{e}5 \text{ m} \\ r_{s2} &= \frac{2Gm_2}{c^2} = \frac{2(6.67430\text{e}-11 \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2})(1.00000\text{e}30 \text{ kg})}{(2.99792\text{e}8 \text{ m s}^{-1})^2} \\ &= 1.48523\text{e}3 \text{ m} \end{aligned}$$

Calculate k values from the r_s/r ratio:

$$\begin{aligned} k_1 &= \frac{r_{s1}}{r_1} = \frac{1.48523\text{e}5\text{ m}}{1.00000\text{e}7\text{ m}} = 1.48523\text{e}-2 \\ k_2 &= \frac{r_{s2}}{r_2} = \frac{1.48523\text{e}3\text{ m}}{1.00000\text{e}8\text{ m}} = 1.48523\text{e}-5 \end{aligned}$$

Calculate GTD via Schwarzschild radius ratio:

$$\begin{aligned} GTD_{s1} &= \sqrt{1 - k_1} = \sqrt{9.851477e-1} = 9.9254606e-1 = 1 - 7.45394e-3 \\ GTD_{s2} &= \sqrt{1 - k_2} = \sqrt{9.999851477e-1} = 9.9999257381e-1 \\ &= 1 - 7.42619e-6 \\ \frac{GTD_{s1}}{GTD_{s2}} &= \frac{9.9254606e-1}{9.9999257381e-1} = 9.9255343e-1 \\ &= 1 - 7.44657e-3 \end{aligned}$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\frac{GTD_{s1}}{GTD_{s2}} = 1 - 7.44657\text{e-}3 \quad (\text{via } r_s)$$

$$\frac{GTD_{d1}}{GTD_{d2}} = 1 - 7.44658\text{e-}3 \quad (\text{via } D_{\text{crit}})$$

4.7 Case 7: (General, unconstrained)

When no constraint is applied between the systems, both mass and radius must be specified independently for each system.

System properties:

	System 1 (S_1)	System 2 (S_2)
m (kg)	1.00000e20	1.00000e30
r (m)	2.00000e-7	1.00000e7

Calculate DTD for each System:

$$\begin{aligned}
 DTD &= \sqrt{k} \quad \text{where} \quad k = m/(rD_{\text{crit}}) \\
 k_1 &= \frac{1.00000e20 \text{ kg}}{(2.00000e-7 \text{ m})(6.73295e26 \text{ kg/m})} = 7.42617e-1 \\
 DTD_1 &= \sqrt{7.42617e-1} = 8.61752e-1 \\
 k_2 &= \frac{1.00000e30 \text{ kg}}{(1.00000e7 \text{ m})(6.73295e26 \text{ kg/m})} = 1.48523e-4 \\
 DTD_2 &= \sqrt{1.48523e-4} = 1.21870e-2 \\
 \frac{DTD_1}{DTD_2} &= \frac{8.61752e-1}{1.21870e-2} = 7.07107e1
 \end{aligned}$$

Calculate the GTD ratio from DTD:

$$\begin{aligned}
 \frac{GTD_{d1}}{GTD_{d2}} &= \frac{\sqrt{1 - DTD_1^2}}{\sqrt{1 - DTD_2^2}} = \frac{\sqrt{1 - (8.61752e-1)^2}}{\sqrt{1 - (1.21870e-2)^2}} \\
 &= \frac{\sqrt{1 - 7.42617e-1}}{\sqrt{1 - 1.48523e-4}} \\
 &= \frac{\sqrt{2.57383e-1}}{\sqrt{9.99851477e-1}} \\
 &= \frac{5.07330e-1}{9.999257356e-1} = 5.07367e-1 \\
 &= 1 - 4.92633e-1
 \end{aligned}$$

Verify by calculating the GTD from the Schwarzschild Radius Ratio:

Schwarzschild radii:

$$\begin{aligned}
 r_{s1} &= \frac{2Gm_1}{c^2} = \frac{2(6.67430e-11 \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2})(1.00000e20 \text{ kg})}{(2.99792e8 \text{ m/s})^2} \\
 &= 1.48523e-7 \text{ m} \\
 r_{s2} &= \frac{2Gm_2}{c^2} = \frac{2(6.67430e-11 \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2})(1.00000e30 \text{ kg})}{(2.99792e8 \text{ m/s})^2} \\
 &= 1.48523e3 \text{ m}
 \end{aligned}$$

Calculate k values from the r_s/r ratio:

$$\begin{aligned}
 k_1 &= \frac{r_{s1}}{r_1} = \frac{1.48523e-7 \text{ m}}{2.00000e-7 \text{ m}} = 7.42616e-1 \\
 k_2 &= \frac{r_{s2}}{r_2} = \frac{1.48523e3 \text{ m}}{1.00000e7 \text{ m}} = 1.48523e-4
 \end{aligned}$$

Calculate GTD via Schwarzschild radius ratio:

$$\begin{aligned}
 GTD_{s1} &= \sqrt{1 - k_1} = \sqrt{2.57384e-1} = 5.07330e-1 = 1 - 4.92670e-1 \\
 GTD_{s2} &= \sqrt{1 - k_2} = \sqrt{9.99851477e-1} = 9.999257356e-1 \\
 &= 1 - 7.42644e-5 \\
 \frac{GTD_{s1}}{GTD_{s2}} &= \frac{5.07330e-1}{9.999257356e-1} = 5.07368e-1 \\
 &= 1 - 4.92632e-1
 \end{aligned}$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\begin{aligned}
 \frac{GTD_{s1}}{GTD_{s2}} &= 1 - 4.92632e-1 \quad (\text{via } r_s) \\
 \frac{GTD_{d1}}{GTD_{d2}} &= 1 - 4.92633e-1 \quad (\text{via } D_{\text{crit}})
 \end{aligned}$$

5 Analysis

This experiment was fascinating. We did not know in advance that it would line up as cleanly as it did, but in retrospect it makes perfect sense, because all the mathematics is already there.

The largest observation we can make is that general and special relativity suffer, significantly, from scale preference: we hold a scale-preferential view of the universe, and it centers on the human scale. We have met this problem before, with space and with other things, but we have never faced it with time. We still treat time as absolute, taking our own time as the preferred clock and measuring everything else against it, even though time is governed by a theory called relativity. We do not even use it that way, and that is not the worst of it. The worst of it is that we do not understand that relativity is relative. Some individuals do, but the physics consensus does not clearly assert the relative nature of relativity, which is why we titled the paper as we did. Relativity is always relative; everything observable is relative, and there is nothing one can observe that is not.

This preference does its damage through a single, reasonable-looking move. A quantity falls off toward infinity and grows very small, and from our own frame we judge it too small to mean anything. That is a fair practical call, but it is not a mathematical one, and it does not make our frame the preferred one. We hear the same move constantly, in a family of interchangeable terms: negligible, almost nothing, infinitesimal, practically nonexistent, weak-field, classical system, flat spacetime, time stoppage, approaching time stoppage, time stops, the domain shifts, it diverges from meaningful results. Most add up to the same thing, and there are no real divergences here; the term is rarely used in its proper sense. The move reduces to two judgment calls made from a distance: that something very large may as well be infinity, and that something very small may as well be zero. Both wreck the arithmetic the moment we treat them as exact at every scale. One question settles it. Is the quantity we discarded large compared with, say, 10^{googol} ? It is not; nothing at our scale is. Measured against a scale that far from our own, the terms we call negligible are enormous. They are negligible to us, never to the systems that carry them.

At our own scale there is little harm in assuming that when we calculate time dilation we calculate it against flat spacetime. But take any flat spacetime you can construct, and you find an infinity of flatter spacetime beyond it. You never reach flat spacetime; however flat it looks, there is always flatter, and it is always the same. Stand at that scale of flatness and you would see the surroundings as curved as they appear to us locally, identical, never a different picture. That is how time works, and how physics works: it is always the same at the observer's scale. Even as you grow it stays the same at your scale, which makes it different between your prior scale and your current one, and that is an accurate statement.

We even get to feel this. We are among the rare things that occupy different scales over a lifetime. As children, adults seemed very large; now, as adults, children seem very small, which is a statement about changing scale over a lifetime. We begin with small, high voices that lower and lower, and we end up large,

bulky beings, quite unlike the children we were, and this changes the experience, spatial and temporal, of the whole body. Yet our cells do not experience time any differently from anyone else's, because they are of roughly the same constituency, size, and density. The brain may scale up or down, but the pace of time in its cells and nerves stays consistent, which is essential for temporal measurement. The body does not have to see what we calculate; it only knows that it runs at a certain pace, and that pace is what is relevant to it. We do not metabolize as fast as a smaller organism, or as slowly as a larger one, on average.

There is more in the analysis. To us, this says there can be no system without mass and radius, which draws two things into question: the photon and the singularity. We hold no personal stake in either, and no reluctance to challenge either if it does not hold up. We raise it as analysis: there is an indication of infinite scale that other views of the cosmos do not necessarily support. It draws the interpretation of the cosmic microwave background into question as well. The CMB is sometimes taken to mean there is a limit to scale, a reading we have heard offered for it, and that reading is not consistent with a theory built on inertia, which requires only that everything in a system have some mass and radius. To suggest there is anything without either is itself a scale-preferential mindset. The same unbounded scale is written into gravitational collapse: the black-hole condition, $m/r = D_{\text{crit}} = 6.73295e26 \text{ kg/m}$, is a single relation between two free quantities, satisfied at every scale, and it bounds neither mass nor radius.

This points out that general relativity is, in fact, the real governing factor, and that it governs without a proliferation of field models. We have not examined the charge-based ones, but Kerr and ordinary black holes are explained equally well by the Schwarzschild metric expanded to three dimensions, which is what happens naturally in an inertial model like this one.

A clear expectation follows. We should see considerable redshift when we look at distant objects that are not themselves deep gravitational wells. Such an object may contain deep wells without being one, and common intuition says it should not redshift; but relative to us it carries significant inertial time dilation, and so it runs at that slower pace of time. This is gravitational: general relativity already accounts for the broad, low-density black holes at the centers of galaxies, and for the observable universe itself. Redshift then has a clean, usable meaning, because it reports the spatial distribution of mass. We read it as a two-dimensional measurement, mass over the mean radius of the circle of sky we are looking at. For a galactic cluster we measure the cluster; for the region around it, we include all of it; and because mass over mean radius gives only the average within that sphere, a smaller sample buys finer resolution. General relativity gives an adequate account of the entirety of the redshift, and from it we read the spatial distribution of mass directly and check it. The test is narrow: does general relativity predict what we see, and does it predict more or less than that? It predicts less, which leaves a real remainder to explain; if it predicted more than we see, the framework would be in trouble, and we leave that possibility open rather than assume it away. These are accessible checks, on well-characterized nearby systems such as Andromeda.

This is a genuine unification of general and special relativity. General relativity is carried by rotational inertia, the moment of inertia; special relativity is carried by ordinary linear inertia, which is mass; and general relativity is born of the moment of inertia. To check it, presume that the inertia of any object or system, about a chosen axis or along a chosen linear dimension, is the pace of time for that whole system, and follow the consequences rationally. Special and general relativity emerge necessarily, which makes it almost inconceivable that they would emerge as they do unless the premise were true. The definition of inertia supports it: inertia, linear or rotational, is the property that translates an applied force into the resulting change in motion, and any two systems with the same moment of inertia, under the same force or torque, undergo the same temporal effect, regardless of how each reached that moment of inertia. Moment of inertia is, then, literally system time. Take that together with inertia's additive and conservational nature, and we have a clean, logical model of causality. None of this changes what we observe or how we perceive it. It is simply a different frame of observation, the system frame. We adopt a system frame and presume it to be the universal frame, but it is not. Whether we call the effect time dilation or time scale, it is the same thing; Einstein would call it indistinguishable, but we need cite nothing, because this is general relativity. To attack it, you would have to attack general relativity.

Here is another consequence, and a real test. Human vision very likely uses a redshift mechanism to perceive color, and if it does not, that points to a serious problem with the theory. For the mechanism to work, the sensing element must sit at a consistent position relative to the incoming light, and the chromophores do: they are mounted to the retina, which is always perpendicular to the incoming light, so each receives light at a consistent angle. The cone-shaped chromophore bends at its middle when a particular wavelength strikes it; bent in the right direction, it covers its own photosensitive portion and the reaction stops, after which it relaxes back to a straight structure and is ready to be excited again. We carry three copies of the same chromophore across three cone sizes, and each shields the angle of light just enough to sample a slightly different range of inertial density, so that from the three together the system triangulates where the incoming signal falls in that range. What the eye is really reading is mass over radius about the axis of observation. Take a chromophore as it faces the plane perpendicular to the incoming light and compute its axial moment of inertia, the quantity we divide by volume or radius to get inertial density, and we get a value on the order of 10^{-43} , a good approximation of what the eye reads. Denser systems, with more mass in a smaller region, have a smaller moment of inertia and a faster pace of time, so as rough magnitudes white and yellow sit near 10^{-43} , red near 10^{-43} , and blue near 10^{-44} , with infrared beyond the red and ultraviolet beyond the blue, and this holds whether the colorant is a solid, a liquid, or a gas.

Chemistry explains these colors by electron transitions, and for the larger pigment molecules that account matches the behavior. The noble gases, though, have no good electron-based explanation and are treated as an anomaly. Inertial relativity

has no anomaly there: it predicts squarely that the mass-to-radius ratio, relative to the observer alone, sets the observed color, regardless of the electron's state, and that is compelling evidence for the mechanism. Inertial density rises as inertia increases over the same volume, up to the DeGerlia Threshold $D_{\text{crit}} = 6.73295\text{e}26\text{ kg/m}$, and the radius at which the electrons orbit the nucleus bears materially on the moment of inertia, enough to shift the time dilation between the observing and observed systems. Neon, for one, turns bluer under compression and redder as the pressure drops. The same reasoning covers the small-particle systems that sit in the ultraviolet and trickle into the visible, such as water, or the daytime blue sky, an illuminated system of objects that blue-shift relative to the eye's chromophore. The decisive test is to predict a compound's ultraviolet absorption spectrum from its geometry alone. Take water, present as H_2O , H_3O^+ , and OH^- : compute each form's mass-to-radius ratio about each of its three axes to get a range of resonance per axis, and weight by proportion to get a representative peak for each three-dimensional orientation. It even opens the door to engineering compounds to a target absorption pattern, made uniform across viewing axes or deliberately non-uniform.

Taken as a whole, the framework proved useful and showed every characteristic we expected of it. The salient finding is about the scale-dilation term. Either it has been rendered entirely out of gravitational time dilation, or, more likely, it has been interpreted away in a human-centric fashion until it carries no meaning. As we suspected, the normalization that drives the value into $[0, 1]$ carries a factor that, taken as negligible, is simply lost, and it should never be taken as negligible with respect to the working of the system in front of us. The framework does make the time-dilation changes clear and intuitive: in the classical example the scale factor came out to 2, exactly as expected, a purely spatial effect, with the time dilation negligible. The real takeaway is DTD against GTD. DTD, $\sqrt{k}$, represents time dilation faithfully in a way the GTD form $\sqrt{1-k}$ does not; strip the "1-" and what remains is a clean and complete framework, of which GTD gives only a shadow. The two carry the same information, but from either one alone we cannot map fully back to the original systems without the actual, real-world parameters of at least one of them.

6 Conclusions

Three conclusions follow from the cases. All three are aspects of a single point: the presumption of a preferred scale. Each is mathematically sound and defensible on its own.

1. **Relativity, general and special, is inherently a two-system comparison, always.** In general relativity the scale factor $k = r_s/r$ is a ratio of two radii, a system's radius and the Schwarzschild radius of its own mass, so the familiar single-system calculation is a two-system comparison in which the second system has been fixed and left unstated. We demonstrate this in the one place hardest to grant it, the ordinary gravitational time-dilation formula everyone reads as a single body, and reinforce it across all eight cases: in each, the relative time dilation computed from DeGerlia inertial density reproduces the standard Schwarzschild result exactly, to

the precision permitted by G . The second system is not an idealization to be discarded; its contribution is real and exact. Special relativity is no exception: it compares a velocity against the maximum attainable velocity, which is itself a comparison, and treating that maximum as an absolute reference is a scale-privileged choice of the same kind.

2. **A privileged perspective on scale (of time or of physical size) is a fallacy that corrupts the mathematics.** Where only one scale is in play the single-system framing gives good results; the damage is in presuming it extends, that a quantity negligible at our scale is negligible for the system itself or at every scale. It is not. The neglected contribution of the second system is $1 - \sqrt{1-k}$, which increases monotonically with $k = D/D_{\text{crit}}$: the error is not incidental, and it compounds without exception as one computes further from one's own scale. Treating flat spacetime as a fixed background is the clearest instance: a benchmark never reached, no flatter at any scale than where one began. It is the silent rounding of the very large to infinity and the very small to zero, and it is exactly there that the mathematics goes wrong.
3. **The relative time dilation between two systems is exactly their DeGerlia comparison.** Across all eight cross-domain cases, the ratio of the systems' gravitational time dilation, computed entirely from their DeGerlia quantities, equals the observed $d\tau/dt$:

$$\frac{d\tau}{dt} = \frac{\sqrt{1 - m_1/(r_1 D_{\text{crit}})}}{\sqrt{1 - m_2/(r_2 D_{\text{crit}})}}.$$

This is the central result the paper demonstrates, and it holds without exception across every case examined.

7 Hypotheses of Explanation

These are items whose explanation we understand and hold to be right. We do not label them *theory* beyond what is warranted, but that restraint is the strength of a hypothesis of explanation, not a retreat from the result. That each was *predicted* by the framework already makes the case strong; what remains genuinely open is only the *meaning* one attaches to some of them. We have not exhaustively demonstrated, numerically and on paper, that every one is consistent in every case with the predictions of general and special relativity, but nothing here requires new observations. No new tests need be run; the observations would not change. The only open question is whether it adds up under this understanding, and it does.

The explanations put forward, each following from the framework and each already consistent with what we observe:

1. *Redshift is inertial time dilation.* Distant systems that are not themselves deep gravitational wells still carry significant inertial time dilation relative to us and therefore redshift. General relativity accounts for the entirety of the effect, and the redshift then reads directly as the spatial distribution of mass, $m/\bar{r}$ over the field of view, a clean and validatable quantity.

2. *Human color vision is a redshift mechanism.* The eye reads mass over radius about the axis of observation. The chromophores' fixed perpendicular mounting to the retina, the cone-size triad, and the inertial-density-to-color mapping all follow from this, and it is a real, falsifiable test of the framework.
3. *A compound's optical resonance is predictable from its geometry.* Mass-to-radius ratios about a molecule's three axes give its resonance ranges and peaks, including the noble-gas behavior (neon bluer under compression, redder under reduced pressure) that electron-transition accounts treat as anomalous.

The value of a hypothesis of explanation is its flexibility: it sets out the account without the overreach of asserting settled theory, and it stays open to test rather than pretending to be closed.

Because the underlying relationship is simultaneously one of inertia and density, the inertia component imparts dimensionality to time. Consequently, although the present exploration is restricted to uniform spherical systems, the framework extends to asymmetrical systems, including large low-density black holes, Kerr black holes⁴, and the observable universe.

By contrast, the unification through inertia and the conclusions above are held to a firmer standard than hypothesis: they follow directly from general relativity and are not offered tentatively.

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Conflicts of Interest

The author declares no competing interests.

Data Availability

The complete source materials for this paper, including all $\text{\LaTeX}$ source files, derivation documents, and revision history, are publicly available at: https://github.com/denverdata/academic_InertialRelativity_STE_cases.

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